

FINAL VS CONST

Final Vs Const In Dart

If you do not want to change the value of a variable, then you can use either final or const in dart.

Example

```
void main() {  
  final finalName = "Final John Doe";  
  const constName = "Const John Doe";  
  
  finalName = "Raj"; // Not Possible  
  constName = "Anu"; // Not Possible  
  
  print("Final name is " + finalName);  
  print("Const name is " + constName);  
}
```

[Show Output](#)

Const In Dart

If you need to calculate value at compile-time, it is a good idea to choose `const` over `final`.  A const variable is a compile-time constant. They must be created from data that can be calculated at compile time. `100+1` is valid const expression but `const date = DateTime.now();`  is not.

What Is Compile Time

When you run code in the dart, it will be `compiled`  into the format that the machine can understand. This time is called compile time. Const value should be known at compile time.

What Is Run Time

Runtime is the time when your compiled code is started running. It generally occurs after the compile time.

Info

Note: If you use `const` inside the class, declare it as `static const`. 

```
const total = 50+50; // Possible
const date = DateTime.now(); // Not Possible
```



Advantage Of Constant

- Improve Performance

Final In Dart

If the value is calculated at runtime, you can choose final for it. For. e.g if you want to calculate date on run time, you can use `final date = DateTime.now();`  but not `const date = DateTime.now();` .

Info

Note: Anything that is unknown at compile time should be `final` over `const`. 

```
final date = DateTime.now(); // Possible
const date = DateTime.now(); // Not Possible
```



When To Use Const

- If you know the value at compile-time, choose `const` for e.g. `const a = 100;` .

When To Use Final

- If you don't know the value at compile-time, choose `final`.

- If you want a network request that can't be changed, choose `final`.
- If you want to get some values from the database, choose `final`.
- If you want to read a local file, choose `final`.

Info

Note: Final variables will have a value known at runtime. Const variables have a value known at compile time. Instance variable can be final but not const.